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Date: 06/05/2026

MONTHLY CHALLENGE

SET D

Duration: 2 hours

III YEAR – SEMESTER 6

Question 1:

1. Problem Title & Link

Title: Minimum Operations to Make Array K-Increasing

2. Problem Statement

You are given:

- An integer array `nums`
- An integer `k`

Definition

An array is **k-increasing** if:

$\text{nums}[i] \leq \text{nums}[i + k]$ for all valid `i`

Goal

You can **modify any element** (replace it with any integer).

- Return the **minimum number of operations** required to make the array k-increasing.

3. Examples

Example 1

Input:

`nums = [5,4,3,2,1]`

`k = 1`

Output:

4

4. Constraints

- $1 \leq \text{nums.length} \leq 10^5$
- $1 \leq \text{nums}[i] \leq 10^9$
- $1 \leq k \leq \text{nums.length}$

Question 2:**1. Problem Title & Link****Title:** Minimum Cost to Make All Subarrays Non-Decreasing**2. Problem Statement**

You are given an integer array `nums`.

You can perform the following operation:

- Choose any index `i` and **increase** `nums[i]` to any value $\geq$ current value
- Cost = amount increased

Condition

After operations, **every subarray** of length ≥ 2 must be **non-decreasing**

For every subarray:

`nums[j] <= nums[j+1]`

Goal

- Return the **minimum total cost** required

3. Examples**Example 1**

Input:

`nums = [3,1,2]`

Output:

2

Explanation:

`[3,3,3] → cost = 2`

4. Constraints

- $1 \leq \text{nums.length} \leq 10^5$
- $1 \leq \text{nums}[i] \leq 10^9$